

Supplementary Material for “Construction of Synchronous Machine Surrogates from Limited Measurements: A Physics-Prior Neural ODE Approach”

Ruizhi Yu, Wei Gu, Shuai Lu, Ke Wang, Suhan Zhang

This document supplements the main text. Section S1 lists the synchronous machine models of the family in full. Section S2 proves Proposition 1 of the main text. Section S3 gives the gradient of the port-only loss. Section S4 gives the numerical differentiation formula and its residual equation. Section S5 lists the parameters of the IEEE 14-bus case study. Equation, section, table, and proposition numbers without the prefix S refer to the main text.

S1. THE SYNCHRONOUS MACHINE MODEL FAMILY

A. The Sauer–Pai Model with GENROU Saturation

The sixth-order Sauer–Pai model [1] with the GENROU saturation [2] reads

$$V_d = V_h \sin(\delta - \theta_h), \quad V_q = V_h \cos(\delta - \theta_h), \quad (\text{S1a})$$

$$\dot{\delta} = \Omega_b(\omega - 1), \quad 2H\dot{\omega} = T_m - T_e - D(\omega - 1), \quad (\text{S1b})$$

$$0 = V_d + R_a I_d - E_d'' - X_q'' I_q, \quad (\text{S1c})$$

$$0 = V_q + R_a I_q - E_q'' + X_d'' I_d, \quad (\text{S1d})$$

$$T_e = E_d'' I_d + E_q'' I_q + (X_q'' - X_d'') I_d I_q, \quad (\text{S1e})$$

$$T_{d0}' \dot{E}_q' = -E_q' - (X_d - X_d') (I_d - \gamma_d \eta_d) + E_{fd} - S_e E_q'', \quad (\text{S1f})$$

$$T_{q0}' \dot{E}_d' = -E_d' + (X_q - X_q') (I_q - \gamma_q \eta_q) - S_e g_{qd} E_d'', \quad (\text{S1g})$$

$$T_{d0}'' \dot{\psi}_{1d} = -\psi_{1d} + E_q' - (X_d' - X_l) I_d, \quad (\text{S1h})$$

$$T_{q0}'' \dot{\psi}_{2q} = -\psi_{2q} - E_d' - (X_q' - X_l) I_q, \quad (\text{S1i})$$

$$E_q'' = a_d E_q' + b_d \psi_{1d}, \quad E_d'' = a_q E_d' - b_q \psi_{2q}, \quad (\text{S1j})$$

$$S_e = \frac{B_s(\psi_a - A_s)^2}{\psi_a}, \quad \psi_a = \sqrt{(E_d'')^2 + (E_q'')^2}. \quad (\text{S1k})$$

Equations (S1a)–(S1e) and (S1j) are the swing and stator equations (3) of the main text, and the symbols there keep their meaning. The rotor equations (S1f)–(S1i) take the transient EMFs E_q' , E_d' and the damper flux linkages ψ_{1d} , ψ_{2q} as rotor states, where E_{fd} is the field voltage, X_d , X_q the synchronous reactances, X_d' , X_q' the transient reactances, X_l the leakage reactance, and T_{d0}' , T_{q0}' , T_{d0}'' , T_{q0}'' the open-circuit time constants. The coupling terms carry the sub-transient leakage fluxes

$$\eta_d = \psi_{1d} + (X_d' - X_l) I_d - E_q', \quad \eta_q = \psi_{2q} + (X_q' - X_l) I_q + E_d', \quad (\text{S2})$$

and the constant coefficients

$$\gamma_d = \frac{X_d' - X_d''}{(X_d' - X_l)^2}, \quad \gamma_q = \frac{X_q' - X_q''}{(X_q' - X_l)^2}. \quad (\text{S3})$$

The EMF map (S1j) has the coefficients

$$\begin{aligned} a_d &= \frac{X_d'' - X_l}{X_d' - X_l}, & b_d &= \frac{X_d' - X_d''}{X_d' - X_l}, \\ a_q &= \frac{X_q'' - X_l}{X_q' - X_l}, & b_q &= \frac{X_q' - X_q''}{X_q' - X_l}. \end{aligned} \quad (\text{S4})$$

The saturation factor (S1k) is a quadratic function of the air-gap flux magnitude ψ_a that vanishes for $\psi_a \leq A_s$, and it enters the q axis through $g_{qd} = (X_q - X_l)/(X_d - X_l)$. Its coefficients A_s and B_s are fitted so that S_e passes through the standard saturation points (1.0, $S_{1.0}$) and (1.2, $S_{1.2}$) [2], and $S_e = 0$ recovers the unsaturated Sauer–Pai model.

B. The Marconato and the Anderson–Fouad Models

The Marconato model [3] takes the sub-transient EMFs themselves as rotor states, E_q' , E_d' , E_q'' , E_d'' , and its rotor equations read

$$T_{d0}' \dot{E}_q' = -E_q' - (X_d - X_d' - \gamma_d^M) I_d + \left(1 - \frac{T_{AA}}{T_{d0}'}\right) E_{fd}, \quad (\text{S5a})$$

$$T_{q0}' \dot{E}_d' = -E_d' + (X_q - X_q' - \gamma_q^M) I_q, \quad (\text{S5b})$$

$$T_{d0}'' \dot{E}_q'' = -E_q'' + E_q' - (X_d' - X_d'' + \gamma_d^M) I_d + \frac{T_{AA}}{T_{d0}''} E_{fd}, \quad (\text{S5c})$$

$$T_{q0}'' \dot{E}_d'' = -E_d'' + E_d' + (X_q' - X_q'' + \gamma_q^M) I_q, \quad (\text{S5d})$$

where T_{AA} is the additional leakage time constant of the d axis and

$$\gamma_d^M = \frac{T_{d0}'' X_d''}{T_{d0}' X_d'} (X_d - X_d'), \quad \gamma_q^M = \frac{T_{q0}'' X_q''}{T_{q0}' X_q'} (X_q - X_q'). \quad (\text{S6})$$

The sub-transient EMFs are states, so the EMF map (S1j) is the identity, and the stator and torque equations (S1c)–(S1e) apply unchanged. The Anderson–Fouad model [3] is (S5) with $\gamma_d^M = \gamma_q^M = 0$ and $T_{AA} = 0$, that is, four cascaded first-order equations without coupling.

C. The Reduced-Order Models

The reduced-order members drop the rotor states whose time constants are separated from the electromechanical time scale [1], [3]. The two-axis model drops the damper windings,

whose time constants T_{d0}'' and T_{q0}'' are of the order of tens of milliseconds, and keeps

$$T_{d0}'\dot{E}_q' = -E_q' - (X_d - X_d')I_d + E_{fd}, \quad (\text{S7a})$$

$$T_{q0}'\dot{E}_d' = -E_d' + (X_q - X_q')I_q, \quad (\text{S7b})$$

$$0 = V_d + R_a I_d - E_d' - X_q' I_q, \quad (\text{S7c})$$

$$0 = V_q + R_a I_q - E_q' + X_d' I_d, \quad (\text{S7d})$$

$$T_e = E_d' I_d + E_q' I_q + (X_q' - X_d') I_d I_q, \quad (\text{S7e})$$

so the transient EMFs act behind the transient reactances. The one-axis model further drops E_d' and keeps

$$T_{d0}'\dot{E}_q' = -E_q' - (X_d - X_d')I_d + E_{fd}, \quad (\text{S8a})$$

$$0 = V_d + R_a I_d - X_q I_q, \quad (\text{S8b})$$

$$0 = V_q + R_a I_q - E_q' + X_d' I_d, \quad (\text{S8c})$$

$$T_e = E_q' I_q + (X_q - X_d') I_d I_q. \quad (\text{S8d})$$

The classical model holds a constant EMF E' behind the transient reactance X_d' , since T_{d0}' is of the order of seconds and the field flux barely changes within a first-swing study, and keeps

$$0 = V_d + R_a I_d - X_d' I_q, \quad (\text{S9a})$$

$$0 = V_q + R_a I_q - E' + X_d' I_d, \quad (\text{S9b})$$

$$T_e = E' I_q. \quad (\text{S9c})$$

In every member the Park transformation (S1a) and the swing equations (S1b) are unchanged.

D. Saturation Conventions

Even for the saturation no consensus exists. Reference [1] distributes one open-circuit saturation function of the air-gap flux over the field and damper windings, which the GENROU form (S1f)–(S1g) and (S1k) follows, while [3] admits a separate saturation function per winding, realized as a three-region piecewise curve or as a polynomial fit of the saturation characteristic. The rotor equations thus vary across references, machine types, and model orders, which is why the main text learns the rotor model instead of presuming one member.

S2. PROOF OF PROPOSITION 1

Proposition 1 of the main text bounds every trajectory of the learned rotor model (7), for any network weights, by (10).

Proof. Write (7) as $\dot{\mathbf{x}}_r = \mathbf{\Lambda} \mathbf{x}_r + \mathbf{w}(t)$ with $\mathbf{\Lambda} = \mathbf{T}^{-1} \mathbf{A}$ and $\mathbf{w} = \mathbf{T}^{-1}(\mathbf{B} \mathbf{u}_r + \mathbf{r}_\theta)$. The output bound of (9) gives $|r_{\theta,i}| \leq c_i$, so $\|\mathbf{w}(t)\|_2 \leq \bar{u} + \|\mathbf{T}^{-1} \mathbf{c}\|_2$ under the input bound $\|\mathbf{T}^{-1} \mathbf{B} \mathbf{u}_r(t)\|_2 \leq \bar{u}$. Every eigenvalue of $\mathbf{\Lambda}$ is negative, so constants $k \geq 1$ and $\lambda > 0$ with $\|e^{\mathbf{\Lambda} t}\|_2 \leq k e^{-\lambda t}$ exist [4]. The variation-of-constants formula gives $\mathbf{x}_r(t) = e^{\mathbf{\Lambda} t} \mathbf{x}_r(0) + \int_0^t e^{\mathbf{\Lambda}(t-s)} \mathbf{w}(s) ds$. Bounding the integrand by $k e^{-\lambda(t-s)} \|\mathbf{w}\|_2$ and integrating yields (10). $\square$

S3. GRADIENT OF THE PORT-ONLY LOSS

The gradient of the loss (14) is obtained by first discretizing the forward simulation and then differentiating the discrete solution, which is exactly what reverse-mode automatic differentiation computes. Let $\mathbf{y} = [\delta, \omega, \mathbf{x}_r^\top]^\top$ collect the surrogate state and let Φ_n denote one step of a fixed-step fourth-order Runge–Kutta scheme applied to the surrogate dynamics under the recorded inputs. The forward propagation of one record of L steps is

$$\mathbf{y}_0 = \iota(\mathbf{X}, \mathbf{P}), \quad \mathbf{y}_{n+1} = \Phi_n(\mathbf{y}_n; \mathbf{P}), \quad n = 0, \dots, L-1, \quad (\text{S10})$$

where ι is the initial-condition map of Section III-C of the main text, $\mathbf{X}$ are the latent initial angles, and $\mathbf{P}$ are the shared parameters. For the backward propagation, write ℓ_n for the contribution of sample n to (14), and $\mathbf{a}_n = \partial \mathcal{L} / \partial \mathbf{y}_n$ for the sensitivity of the loss to the state at sample n . The sensitivity obeys the reverse recursion

$$\mathbf{a}_L = \frac{\partial \ell_L}{\partial \mathbf{y}_L}, \quad \mathbf{a}_n = \frac{\partial \ell_n}{\partial \mathbf{y}_n} + \left(\frac{\partial \Phi_n}{\partial \mathbf{y}_n} \right)^\top \mathbf{a}_{n+1} \quad (\text{S11})$$

for $n = L-1, \dots, 0$. The gradient with respect to the learnable variables $\boldsymbol{\vartheta} = (\mathbf{X}, \mathbf{P})$ then assembles from three terms,

$$\frac{d\mathcal{L}}{d\boldsymbol{\vartheta}} = \sum_{n=0}^{L-1} \left(\frac{\partial \Phi_n}{\partial \boldsymbol{\vartheta}} \right)^\top \mathbf{a}_{n+1} + \sum_{n=0}^L \frac{\partial \ell_n}{\partial \boldsymbol{\vartheta}} + \left(\frac{\partial \mathbf{y}_0}{\partial \boldsymbol{\vartheta}} \right)^\top \mathbf{a}_0, \quad (\text{S12})$$

summed over the records. The first term carries the influence of $\mathbf{P}$ on the dynamics, and the second its direct influence on the predicted current through the stator equations (3c)–(3d). The third carries the influence of the initial condition. It is the only term that reaches $\mathbf{X}$, and it adds to the gradient that the first two terms already give $\mathbf{P}$.

S4. THE NUMERICAL DIFFERENTIATION FORMULA

Write $\mathbf{z} = [\mathbf{x}^\top, \mathbf{y}^\top]^\top$ for the full unknown vector of the DAE (1), $\mathbf{F} = [\mathbf{f}^\top, \mathbf{g}^\top]^\top$ for its right-hand side, and $\mathbf{M} = \text{diag}(\mathbf{I}, 0)$ for the mass matrix, so that the DAE is $\mathbf{M} \dot{\mathbf{z}} = \mathbf{F}(t, \mathbf{z})$. The step from t_n to t_{n+1} is taken by the numerical differentiation formula (NDF) of order k [5],

$$\mathbf{M} \left(\sum_{m=1}^k \frac{1}{m} \nabla^m \mathbf{z}_{n+1} - \kappa \gamma_k \nabla^{k+1} \mathbf{z}_{n+1} \right) = h \mathbf{F}(t_{n+1}, \mathbf{z}_{n+1}), \quad (\text{S13})$$

where h is the step size, ∇ is the backward difference operator with $\nabla^0 \mathbf{z}_n = \mathbf{z}_n$ and $\nabla^{j+1} \mathbf{z}_n = \nabla^j \mathbf{z}_n - \nabla^j \mathbf{z}_{n-1}$, $\gamma_k = \sum_{j=1}^k 1/j$, and κ is the damping coefficient that lowers the leading truncation term. With the predictor $\mathbf{z}_{n+1}^{(0)} = \sum_{m=0}^k \nabla^m \mathbf{z}_n$, equation (S13) becomes the residual equation (15) of the main text,

$$\mathbf{G}(\mathbf{z}) = \mathbf{M}(\mathbf{z} - \mathbf{z}_{n+1}^{(0)} + \boldsymbol{\Psi}) - \alpha_k \mathbf{F}(t_{n+1}, \mathbf{z}) = \mathbf{0},$$

$$\alpha_k = \frac{h}{(1 - \kappa) \gamma_k}, \quad \boldsymbol{\Psi} = \frac{1}{(1 - \kappa) \gamma_k} \sum_{m=1}^k \gamma_m \nabla^m \mathbf{z}_n, \quad (\text{S14})$$

which is one nonlinear system per step. The simplified Newton iteration that solves it is (16) of the main text.

S5. PARAMETERS OF THE IEEE 14-BUS CASE STUDY

The five members of the family in Section V-A of the main text share the parameters of the machine at bus 2 of the IEEE 14-bus system [3], namely $H = 3.924$ s, $D = 1.2$, $R_a = 0.0052$, $X_d = 1.75$, $X_q = 1.633$, $X'_d = 0.308$, $X'_q = 0.6$, $X''_d = X''_q = 0.217$, and $X_l = 0$ in per unit, and $T'_{d0} = 6.1$, $T'_{q0} = 0.3$, $T''_{d0} = 0.04$, and $T''_{q0} = 0.099$ s. The GENROU setting adds the saturation factors $S_{1.0} = 0.27$ and $S_{1.2} = 1.14$. Every learnable parameter of the surrogate starts from these exact values.

REFERENCES

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